

SUSHEEL R N

Data Scientist (NLP | GenAI | Computer Vision)

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SKILLS

Programming Languages:

Python, JavaScript, c#, HTML

Machine Learning & Deep Learning:

Scikit-Learn, TensorFlow, PyTorch, Keras

Natural Language Processing (NLP):

Hugging Face, OpenAI, LangChain, LangGraph, CrewAI, AutoGen

Computer Vision (CV):

OpenCV, Ultralytics, YOLO, OpenGL, MediaPipe

AR/VR:

Unity, Vuforia, Oculus VR

Data Engineering & MLOps:

Apache Kafka, Apache Airflow, FastAPI, Flask, Django, Docker

Databases:

PostgreSQL, MySQL, SQLite, MongoDB

PROFESSIONAL SUMMARY

Results driven Data Scientist with 5+ years of expertise in NLP, Generative AI, and Computer Vision. Proven track record of building and deploying AI solutions that enhance efficiency, automation, and decision-making. Skilled in designing scalable ML/DL/GenAI models that solve real-world challenges with greater business impact.

GenAI Projects

Lead Prediction System: Led development of an automated pipeline to identify potential leads from user-configured websites using Python, web scraping, and LLMs. Integrated Airflow for weekly alerts, reducing manual monitoring and accelerating lead discovery.

Risk Prediction for Tenders: Built a risk detection model using existing libraries, Advanced RAG, and LLM agents to analyze historical tender data, improving early risk identification and enabling faster, data-driven decisions.

RFQ Automation Platform: Architected an end-to-end system to automate procurement identification and RFQ generation, extracting and summarizing procurement information across all pages using LLM- and agent-based pipelines. Integrated a chatbot interface to answer user queries on tender projects, reducing processing effort and turnaround time.

LLM-Powered Search Engine: Developed an intelligent chatbot for 15,000+ unstructured documents, designed for senior decision-makers to efficiently retrieve information from past tenders. Leveraged Databricks, custom section detection, and intelligent chunking to improve search accuracy and relevance.

AI Auto-Completion System: Built a GPT-powered feature that recommends responses dynamically as users type, improving accuracy, speeding up drafting, and assisting users in generating or modifying answers directly in the answer box.

RAG-Based Response Assistant: Implemented GPT-powered features for first-draft generation, summarization, proofreading, and readability enhancement using RAG pipelines, enabling users to draft or modify responses in real time, resulting in faster and higher-quality RFQ response creation.

Enhanced Recommendation Engine: Upgraded the recommendation pipeline by integrating Ada embeddings with larger embedding dimensions than BERT, achieving 60% improvement in accuracy and relevance for RFP suggestions.

NLP Projects

Document Summarization: Developed a hybrid summarization pipeline combining BERT-based extractive techniques with GPT models to generate concise, high-quality summaries that capture critical content efficiently.

Document Analysis & Requirement Detection: Built an NLP-driven system using open-source libraries to extract key information (dates, sections, locations, organizations, job titles, laws, monetary values, names, percentages, contact details, URLs, domain-specific keywords) and fine-tuned BERT/T5 for sentence classification to automate question/statement identification, achieving 85% efficiency improvement.

Answer Snippet Extraction: Designed a SQuAD fine-tuned BERT model to retrieve the most contextually relevant portions of answers from large unstructured documents, eliminating manual navigation and speeding up information retrieval.

Important Keyword Extraction: Implemented an algorithm leveraging SpaCy, NLTK, TextRank, RAKE, YAKE, KeyBERT to extract meaningful, domain-specific keywords, enhancing search functionality and retrieval accuracy.

Recommendation Engine: Designed an ML-powered retrieval pipeline using BERT vector embeddings and similarity search to suggest the most relevant questions from a database, increasing efficiency in responding to RFPs.

Search & Logging:
Elasticsearch, Logstash,
Kibana

Cloud & DevOps:
AWS (EC2, S3, Lambda,
Bedrock, Boto3), Azure
(AzureML, Blob, Compute, AI
Foundry)

Deployment & Inference:
DeepStream, NVIDIA Triton,
Roboflow

Web Scraping & Automation:
BeautifulSoup, Selenium,
Scrapy

Tools / IDEs:
Jupyter, PyCharm, Spyder, VS
Code, Google Colab,
Anaconda, Cloud VMs

Edge & Embedded Devices:
NVIDIA Jetson, Arduino,
NodeMCU, Raspberry Pi

Model Optimization:
ONNX, TensorRT, OpenVINO

Testing Frameworks:
PyTest, Unittest

RECOGNITIONS

Awarded as Best Outgoing
Student of 2020 with Dr. R
Subbayyan Gold Medal from
BIT, Sathy.

Trailblazer award 2023 from
Responsive, Coimbatore.

EDUCATION

M.Tech AI and ML – BITS Pilani,
India.
2025 – present

B.E ECE – BIT Sathy, India.
2016 – 2020

Computer Vision Projects

Person Footfall Prediction in Retail Stores: Developed a custom person re-identification module with OSNet integrated with YOLO/RT-DETR person detection to reduce identity switches and improve tracking continuity. Used the system to measure footfall, identify high-traffic areas, and predict customer product interest, enhancing retail layout, marketing strategies, and customer analytics.

Vehicle Detection & License Plate Tracking: Developed vehicle detection pipelines using YOLOv8. Combined with LLM-powered OCR (e.g., Gemini) to detect and log license plates, enabling automated monitoring, security, and historical analytics in industrial environments.

Gender & Customer Analytics: Built models to track person in/out counts and gender classification from CCTV footage, providing insights into customer demographics and supporting retail marketing and sales strategies.

Face Recognition for Attendance Monitoring: Developed a face recognition-based attendance system using YOLO-Face for detection and a custom ResNet-based feature extractor, enabling accurate workforce check-ins.

Vehicle Speed Detection in Industrial Environments: Deployed a real-time vehicle speed detection system, calibrating camera distance and perspective using homography and object detection for reliable speed estimation.

PPE Detection for Industrial Safety: Built an end-to-end pipeline for detecting helmets, vests, and gloves. Curated and annotated datasets using Labellmg and Roboflow and trained YOLOv8m, ensuring compliance monitoring in safety-critical zones.

Industrial Hand Monitoring via Pose Estimation: Designed a rule-based system using MediaPipe and YOLO-Pose to monitor hand positions near machinery, enabling early detection of unsafe gestures and accident prevention.

Action Recognition for Retail Theft Detection: Built LSTM-based temporal models to identify suspicious behaviors, contributing to automated security alert systems.

Video Question-Answering (R&D) using NVIDIA Metropolis: Developed a real-time Video QA system leveraging the NVIDIA Metropolis platform, enabling users to ask questions about live video streams for industrial and retail analytics.

Augmented Reality & Virtual Reality Projects

AR Library from Scratch (Python): Built a custom augmented reality pipeline from scratch using Python for image overlay on targeted markers, supporting both static images and real-time video input.

AR Virtual ID (Unity + Vuforia): Developed an AR-based virtual identification system using Vuforia in Unity, enabling real-time camera input to display virtual information overlays within a mobile application.

VR Real Estate Walkthrough (Oculus): Created an interactive virtual reality real-estate walkthrough experience for Oculus devices, allowing users to explore properties in an immersive, 3D environment.

Unity Shooting Game: Developed a space shooter game in Unity using C#, implementing gameplay mechanics, player controls, enemy AI, scoring, and visual effects.

EXPERIENCE

- Data Scientist at LTIMindTree Jan 2024 – Present
- ML Engineer at Responsive Mar 2023 – Dec 2023
- Associate ML Engineer at Responsive Nov 2020 – Feb 2023
- Student Intern at Responsive May 2020 – Nov 2020